

Phase 8: 2026 Rookie Prediction

The purpose of this phase is to use the best-performing model from the evaluation phase to generate forward-looking predictions for the 2026 rookie class.

Based on prior model comparison, Random Forest was selected because it produced the strongest overall performance across both regression and classification tasks.

Two predictions are generated:

- 1. Predicted rookie season total PPR fantasy points
- 2. Predicted fantasy relevance classification

These predictions are not meant to be perfect projections. They represent how the trained model interprets the 2026 class based on historical patterns from previous first-round offensive skill position rookies.

```
In [1]: import pandas as pd
import numpy as np

from sklearn.compose import ColumnTransformer
from sklearn.preprocessing import StandardScaler, OneHotEncoder
from sklearn.pipeline import Pipeline
from sklearn.ensemble import RandomForestRegressor, RandomForestClassifier

from sklearn.metrics import mean_absolute_error, mean_squared_error, accuracy_score, precision_score, recall_score, f1_score, roc_auc_score

In [2]: # Load Excel File

file_path = "Fantasy-Lord-Rookie-ML-Dataset.xlsx"

draft_df = pd.read_excel(file_path, sheet_name="draft_and_player_data")
target_df = pd.read_excel(file_path, sheet_name="fantasy_targets")
master_df = pd.read_excel(file_path, sheet_name="master_dataset")

print(draft_df.shape)
print(target_df.shape)
print(master_df.shape)

(164, 14)
(164, 7)
(164, 17)

In [3]: # Check your Columns

master_df.columns.tolist()
```

```
Out[3]: ['player_id',
        'player_name',
        'season',
        'position',
        'nfl_team',
        'draft_round',
        'draft_pick_overall',
        'age_on_draft_day',
        'height_inches',
        'weight_lbs',
        'college',
        'college_games_played',
        'career_yards',
        'career_touchdowns',
        'rookie_total_ppr_points',
        'season_positional_finish',
        'fantasy_relevant_binary']
```

```
In [4]: # Split Completed Seasons vs 2026 Predictions
```

```
completed_df = master_df[master_df["season"] <= 2025].copy()
predict_2026_df = master_df[master_df["season"] == 2026].copy()

print("Completed seasons:", completed_df.shape)
print("2026 prediction class:", predict_2026_df.shape)
```

Completed seasons: (154, 17)

2026 prediction class: (10, 17)

```
In [5]: # Remove Rows Missing Target from Completed Data
```

```
completed_df = completed_df.dropna(subset=[
    "rookie_total_ppr_points",
    "fantasy_relevant_binary"
])

print(completed_df.shape)
```

(154, 17)

```
In [6]: # Define Features and Targets
```

```
feature_cols = [
    "draft_pick_overall",
    "age_on_draft_day",
    "height_inches",
    "weight_lbs",
    "college_games_played",
    "career_yards",
    "career_touchdowns",
    "position"
]

X = completed_df[feature_cols]

y_reg = completed_df["rookie_total_ppr_points"]
y_class = completed_df["fantasy_relevant_binary"].astype(int)
```

```
X_2026 = predict_2026_df[feature_cols]
```

```
In [7]: # Build Preprocessing Pipelines

numeric_features = [
    "draft_pick_overall",
    "age_on_draft_day",
    "height_inches",
    "weight_lbs",
    "college_games_played",
    "career_yards",
    "career_touchdowns"
]

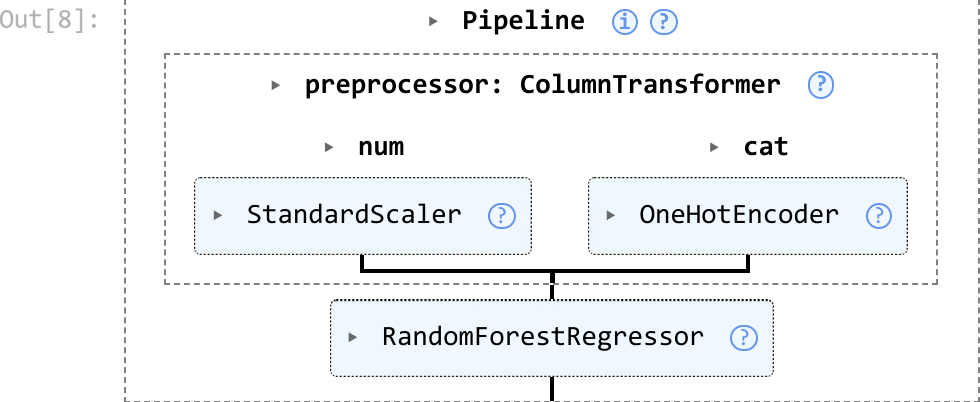
categorical_features = ["position"]

preprocessor = ColumnTransformer(
    transformers=[
        ("num", StandardScaler(), numeric_features),
        ("cat", OneHotEncoder(handle_unknown="ignore"), categorical_features)
    ]
)
```

```
In [8]: # Train Final Random Forest Regression Model
```

```
rf_reg_final = Pipeline(steps=[
    ("preprocessor", preprocessor),
    ("model", RandomForestRegressor(
        n_estimators=500,
        random_state=88,
        max_depth=None,
        min_samples_split=2,
        min_samples_leaf=1
    ))
])

rf_reg_final.fit(X, y_reg)
```



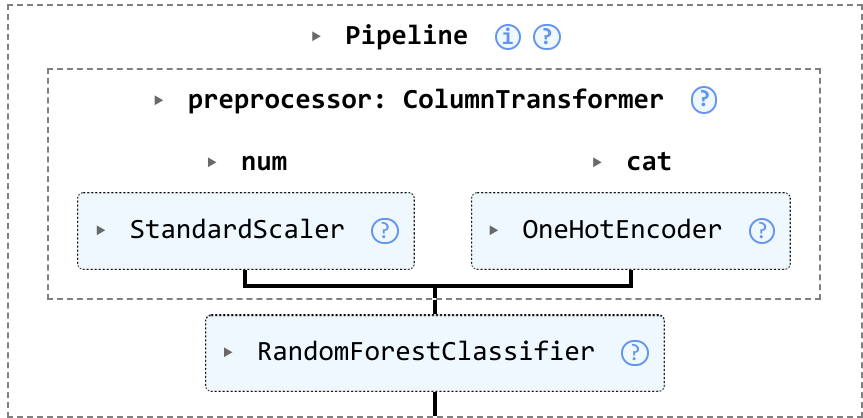
```
In [9]: # Train Final Random Forest Classification Model
```

```
rf_clf_final = Pipeline(steps=[
```

```
("preprocessor", preprocessor),
("model", RandomForestClassifier(
    n_estimators=500,
    random_state=88,
    max_depth=None,
    min_samples_split=2,
    min_samples_leaf=1,
    class_weight="balanced"
))
])

rf_clf_final.fit(X, y_class)
```

Out[9]:



In [10]: *# Generate 2026 Predictions*

```
predict_2026_df["predicted_rookie_ppr_points"] = rf_reg_final.predict(X_2026)

predict_2026_df["predicted_fantasy_relevant_binary"] = rf_clf_final.predict(X_2026)

predict_2026_df["fantasy_relevance_probability"] = rf_clf_final.predict_proba(X_2026)[:, 1]
```

In [11]: *# Create Clean Prediciton Table*

```
prediction_results = predict_2026_df[[
    "player_id",
    "player_name",
    "season",
    "position",
    "draft_pick_overall",
    "predicted_rookie_ppr_points",
    "predicted_fantasy_relevant_binary",
    "fantasy_relevance_probability"
]].copy()

prediction_results["predicted_rookie_ppr_points"] = prediction_results["predicted_rookie_ppr_points"].round(1)
prediction_results["fantasy_relevance_probability"] = (prediction_results["fantasy_relevance_probability"] * 100).round(1)

prediction_results = prediction_results.sort_values(
    by="predicted_rookie_ppr_points",
    ascending=False
)
```

prediction_results

Out[11]:

	player_id	player_name	season	position	draft_pick_overall	predicted_rookie_ppr_points	predicted_fantasy_relevant_binary	fantasy_relevance_probability
155	rb_2026_love	Jeremiyah Love	2026	RB	3	279.8	1	71.8
154	qb_2026_mendoza	Fernando Mendoza	2026	QB	1	223.4	0	6.2
156	wr_2026_tate	Carnell Tate	2026	WR	4	197.5	1	52.6
161	wr_2026_concepcion	KC Concepcion	2026	WR	24	190.3	0	46.4
158	qb_2026_simpson	Ty Simpson	2026	QB	13	170.7	0	39.8
159	te_2026_sadiq	Kenyon Sadiq	2026	TE	16	142.8	0	45.2
160	wr_2026_lemon	Makai Lemon	2026	WR	20	132.5	0	20.0
163	rb_2026_price	Jadarian Price	2026	RB	32	131.1	0	45.2
157	wr_2026_tyson	Jordyn Tyson	2026	WR	8	120.8	1	54.4
162	wr_2026_cooper-jr.	Omar Cooper-Jr.	2026	WR	30	90.3	0	27.8

In [12]:

```
# Add Plain-English Relevance Labels

prediction_results["prediction_label"] = prediction_results["predicted_fantasy_relevant_binary"].map({
    1: "Fantasy Relevant",
    0: "Not Fantasy Relevant"
})

prediction_results
```

Out[12]:

	player_id	player_name	season	position	draft_pick_overall	predicted_rookie_ppr_points	predicted_fantasy_relevant_binary	fantasy_relevance_probability	prediction_label
155	rb_2026_love	Jeremiyah Love	2026	RB	3	279.8	1	71.8	Fantasy Relevant
154	qb_2026_mendoza	Fernando Mendoza	2026	QB	1	223.4	0	6.2	Not Fantasy Relevant
156	wr_2026_tate	Carnell Tate	2026	WR	4	197.5	1	52.6	Fantasy Relevant
161	wr_2026_concepcion	KC Concepcion	2026	WR	24	190.3	0	46.4	Not Fantasy Relevant
158	qb_2026_simpson	Ty Simpson	2026	QB	13	170.7	0	39.8	Not Fantasy Relevant
159	te_2026_sadiq	Kenyon Sadiq	2026	TE	16	142.8	0	45.2	Not Fantasy Relevant
160	wr_2026_lemon	Makai Lemon	2026	WR	20	132.5	0	20.0	Not Fantasy Relevant
163	rb_2026_price	Jadarian Price	2026	RB	32	131.1	0	45.2	Not Fantasy Relevant
157	wr_2026_tyson	Jordyn Tyson	2026	WR	8	120.8	1	54.4	Fantasy Relevant
162	wr_2026_cooper-jr.	Omar Cooper-Jr.	2026	WR	30	90.3	0	27.8	Not Fantasy Relevant

In [13]:

```
# Save Predictions to CSV
```

```
prediction_results.to_csv("data/processed/2026_rookie_predictions.csv", index=False)
```

In [14]: *# Save Predicitons to Excel*

```
prediction_results.to_excel("2026_rookie_predictions.xlsx", index=False)
```